

Solutions to Chapter 2 Exercises:

1. Bernoulli Distribution

- Failure probability: $1 - p = 0.18$.
- Expected successful transmissions: $p = 0.82$.

2. Binomial Distribution

- $P(X = 18) = \binom{20}{18} 0.9^{18} 0.1^2$.
- Mean: $np = 18$.
- Variance: $np(1 - p) = 1.8$.

3. Geometric Distribution

- $P(X \geq 4) = (1 - p)^3 = 0.7^3 = 0.343$.
- $E[X] = 1/p = 3.33$.

4. Poisson Distribution

- $P(N = 10) = \frac{12^{10} e^{-12}}{10!}$.
- $P(N \geq 15) = 1 - F(14)$.
- Expected arrivals in 0.5 s: 6.

5. Exponential Distribution

- $\lambda = 1/0.05 = 20$.
- $P(T < 0.02) = 1 - e^{-20(0.02)}$.
- Mean: $1/\lambda = 0.05\text{s}$.

6. Uniform Distribution

- $P(T < 4) = 4/10 = 0.4$.
- $E[T] = 5$.
- Variance = $\frac{(10-0)^2}{12} = 8.33$.

7. Gaussian Distribution

- Standardize: $z = \pm 1$.
- Probability: $\Phi(1) - \Phi(-1) = 0.6826$.
- About 68% lie within 1 std.

8. t-Distribution

- 95% CI: $\bar{x} \pm t_{0.025,11} \frac{s}{\sqrt{n}}$.
- t used because $n < 30$ and σ unknown.

9. Gamma Distribution

- Mean = $k\theta = 6$.
- Variance = $k\theta^2 = 12$.
- $P(T < 2)$ via Gamma CDF table.

10. Power-Law Distribution

- Heavy-tailed $\Rightarrow$ non-negligible extreme values.
- $P(X > 10) = 10^{-2.5}$.
- Common due to heterogeneous traffic.

11. Laplace Distribution

- Mean = 0.
- Variance = $2b^2 = 18$.
- $P(|X| > 5) = e^{-5/3}$.
- Laplace used for differential privacy because of sharp peak + heavy tails.

12. Mixture of Gaussians

- Expected RSSI

$$E[X] = 0.6(-50) + 0.4(-55) = -52$$

- PDF is bimodal, wider.
- Mixture fits networks with heterogeneous APs.

13. Maximum Likelihood Estimation

- Likelihood: $L(\lambda) = \lambda^n e^{-\lambda \sum x_i}$.
- MLE: $\hat{\lambda} = \frac{n}{\sum x_i}$.
- Numerically: $\sum x_i = 100$, so $\hat{\lambda} = 0.1$.

14. Markov Chain

- Solve $\pi P = \pi$.
- Example steady state (compute fully if needed).
- Two-step transition: use P^2 .

15. Chebyshev's Inequality

- $P(|X - 20| > 10) \leq \frac{25}{100} = 0.25$.
- Gaussian bound would be much tighter.

16. Monte Carlo Approximation

- Mean $\approx$ theoretical mean.
- Larger samples $\rightarrow$ smaller error.

17. Entropy

$$H = -\sum p_i \log_2 p_i = 1.742 \text{ bits}$$

- Interprets average bits/symbol.
- Huffman approaches but never goes below entropy.

18. Gradient Descent

- $f'(x) = 2(x - 3)$.
- Updates:
 - $x_1 = 10 - 0.1 \cdot 14 = 8.6$
 - $x_2 = 8.6 - 0.1 \cdot 11.2 = 7.48$
 - $x_3 = 7.48 - 0.1 \cdot 8.96 = 6.584$
- Sequence moves toward 3.